

BDA400 – Assignment 6: Technical Analysis using R, Visualization Phase

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Assignment: Technical Analysis using R, Visualization Phase (15%)

GitHub repository: <https://github.com/cdimmk1985-crypto/TechnicalAnalysis/tree/main/A06>

Project Description

An interactive portfolio dashboard built with R Shiny. It fetches live daily stock data from Yahoo Finance using quantmod, lets the user pick a stock symbol, date range, and time frame (Daily/Weekly/Monthly), and visualizes it as a line, candlestick, or area chart using ggplot2. Moving Averages are overlaid directly on the price chart, while RSI and MACD are shown as their own panels underneath (toggled on/off independently), since their scales don't match the price chart. A moving-average-crossover trading rule generates Buy/Sell/Hold signals, and the actual crossover points are annotated on the chart with triangle markers plus a signal log table.

How to Run

1. Open RStudio and set the working directory to the A06 folder.
2. Install the required packages if you don't already have them: `install.packages(c("shiny", "ggplot2", "quantmod", "TTR"))`
3. Run: `shiny::runApp("app.R")`
4. The app needs an internet connection since it fetches live data from Yahoo Finance via quantmod's `getSymbols()`.

Design Decisions

- **Indicator overlay approach:** The assignment's own skeleton code shows adding indicators as layers directly on the price chart (`geom_ma()`, `geom_rsi()`, `geom_macd()` - these aren't real ggplot2 functions, just placeholder pseudocode). Doing that literally for RSI/MACD would make the price line unreadable, since RSI ranges 0-100 and MACD is a small positive/negative number, both totally different scales from a stock's dollar price. So Moving Averages (same price scale) are overlaid directly on the price chart, while RSI and MACD get their own stacked sub-panels below - this is how real trading platforms do it, and it still satisfies turning each indicator's panel on/off independently via its own checkbox.
- **Trading rule:** Simple moving average crossover (short-term MA crossing above/below long-term MA), as suggested as an example in the assignment instructions. Both MA periods are adjustable via sliders (default 20/50 days).
- **Signal annotations:** Rather than labeling every single day (which would be unreadable clutter), only the actual crossover points - where the signal changes from the previous day - are marked on the chart with green Buy / red Sell triangles, plus listed in a signal log table below the chart.

Verification

The data-aggregation and signal-generation logic (`core_logic.R`) was pulled out of the Shiny app and unit-tested independently against real 2025 daily data for AAPL (the same dataset from Assignments 2 and 5): weekly/monthly aggregation was checked for correct row counts, the moving-average-crossover signal was cross-checked against an independent manual recalculation (matched exactly), and RSI/MACD outputs were confirmed to fall within their expected ranges. Every chart type (Line/Candlestick/Area) was also test-rendered with and without the Moving Average overlay and signal annotations using real data, confirming the annotated Buy/Sell triangles land exactly where the short and long moving averages actually cross on the real AAPL price history.